

Shiv Nadar University Chennai
CS3807 – Deep Learning Laboratory
Experiment 7
End-to-End Study of Autoencoders, Convolutional
Autoencoders, Denoising Autoencoders and Variational Autoencoders

Degree & Branch	B.Tech Artificial Intelligence & Data Science	Semester V
Subject Code & Name	CS3807 – Deep Learning Laboratory	AY: 2026–27

1. Objective

The objective of this experiment is to develop an end-to-end understanding of autoencoders and their variants for image representation, reconstruction, denoising and generative modeling. The experiment progresses from a fully connected autoencoder to a Convolutional Autoencoder (CAE), introduces controlled image corruption for denoising, and finally develops a Variational Autoencoder (VAE). The learned representations are visualized and the reconstruction quality is evaluated using MSE, MAE and SSIM.

The same MNIST image dataset and a controlled experimental protocol were used throughout the main experiment wherever applicable.

2. Learning Outcomes

After completing this experiment, the following outcomes were demonstrated:

- explain the encoder–latent–decoder structure of an autoencoder;
- prepare image data for reconstruction-based learning;
- implement a fully connected autoencoder;
- implement a Convolutional Autoencoder for image reconstruction;
- introduce controlled Gaussian and salt-and-pepper noise and build a denoising autoencoder;
- evaluate reconstruction using MSE, MAE and SSIM;
- visualize and interpret latent representations;
- explain the probabilistic latent representation of a VAE;
- implement the reparameterization trick;
- calculate and interpret reconstruction and KL-divergence losses;
- generate new images by sampling from a learned latent space; and
- compare deterministic and probabilistic latent representations.

3. Dataset and Experimental Setup

Dataset: MNIST Handwritten Digit Dataset

MNIST contains grayscale handwritten digit images from 0 to 9. Each image has spatial dimensions

$$28 \times 28 \times 1.$$

A subset of 10,000 training images and 2,000 test images was used, as specified in the laboratory protocol. Pixel values were normalized from $[0, 255]$ to $[0, 1]$.

For the fully connected autoencoder, the images were flattened:

$$28 \times 28 = 784, \quad x \in \mathbb{R}^{784}.$$

For convolutional models, the spatial representation

$$x \in \mathbb{R}^{28 \times 28 \times 1}$$

was retained. Labels were not used as reconstruction targets; they were used only for visualization of the VAE latent space and high-error samples.

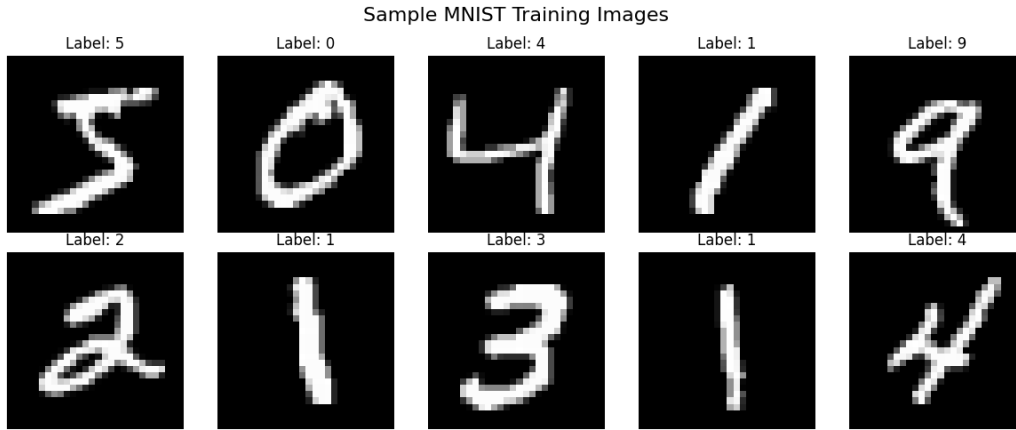


Figure 1: Representative MNIST test images used in the experiment.

4. Overall Experimental Pipeline

The implemented pipeline follows:

MNIST $\rightarrow$ Normalization $\rightarrow$ Encoder $\rightarrow$ Latent Representation $\rightarrow$ Decoder $\rightarrow$ Reconstructed Image $\rightarrow$ Evaluation

Four related studies were completed:

1. Fully Connected Autoencoder;
2. Convolutional Autoencoder;
3. Denoising Convolutional Autoencoder;
4. Variational Autoencoder.

5. Fully Connected Autoencoder

5.1 Architecture and Configuration

The fully connected autoencoder used the architecture

$$784 \rightarrow 128 \rightarrow 32 \rightarrow 16 \rightarrow 32 \rightarrow 128 \rightarrow 784.$$

ReLU activation was used in hidden layers and sigmoid activation in the output layer. The configuration was Adam optimization with learning rate 10^{-3} , batch size 128, 20 epochs and binary cross-entropy reconstruction loss.

The latent representation is a deterministic vector of dimension 16:

$$z = f_{\theta}(x), \quad \hat{x} = g_{\phi}(z).$$

5.2 Training and Validation Loss

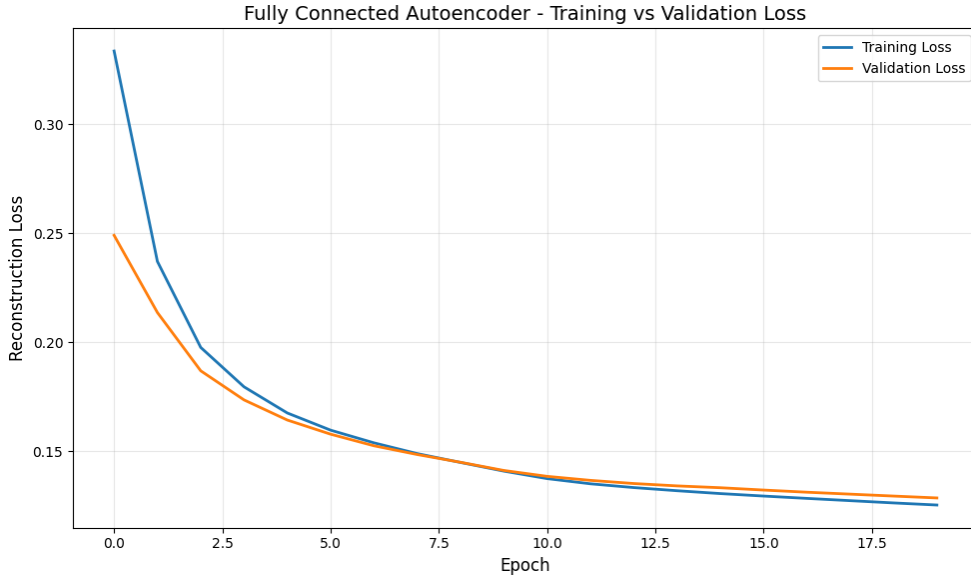


Figure 2: Training and validation reconstruction loss for the fully connected autoencoder.

The training loss decreased from approximately 0.333636 to 0.125405, while validation loss decreased from approximately 0.249100 to 0.128652. The two curves remained relatively close near the end of training. The recorded training time was 23.57 seconds.

Inference: The loss curves show convergence during the 20 epochs. Since the final training and validation losses are close, the recorded run does not show a strong divergence characteristic of severe overfitting.

5.3 Original versus Reconstructed Images

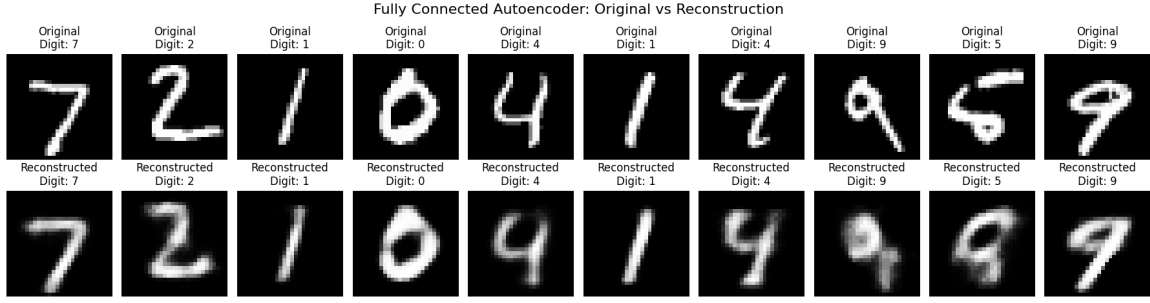


Figure 3: Original MNIST images compared with fully connected autoencoder reconstructions.

The reconstruction preserves the main digit shapes, while some fine stroke details are lost and some regions appear smoother or blurred. Digits with simple, well-defined structures are generally easier to reconstruct than samples containing unusual stroke patterns.

6. Reconstruction Metrics

For a test image x and reconstruction $\hat{x}$, the reconstruction metrics used were

$$\text{MSE} = \frac{1}{N} \sum_{i=1}^N (x_i - \hat{x}_i)^2,$$

$$\text{MAE} = \frac{1}{N} \sum_{i=1}^N |x_i - \hat{x}_i|,$$

and SSIM was computed between each original and reconstructed image and then averaged.

The final test-set values from the executed experiment were:

Metric	FC Autoencoder
Test MSE	0.022515
Test MAE	0.059259
Mean SSIM	0.734634
Parameters	211,040

7. Convolutional Autoencoder

7.1 Architecture

The convolutional autoencoder preserved spatial locality using convolution and pooling operations. The implemented structure follows the laboratory architecture:

$$\text{Input}(28, 28, 1) \rightarrow \text{Conv2D}(32) \rightarrow \text{MaxPool} \rightarrow \text{Conv2D}(64) \rightarrow \text{MaxPool} \rightarrow \text{Conv2D}(64)$$

followed by upsampling and convolutional decoding to a sigmoid output of size $28 \times 28 \times 1$.

The model contained 74,497 parameters.

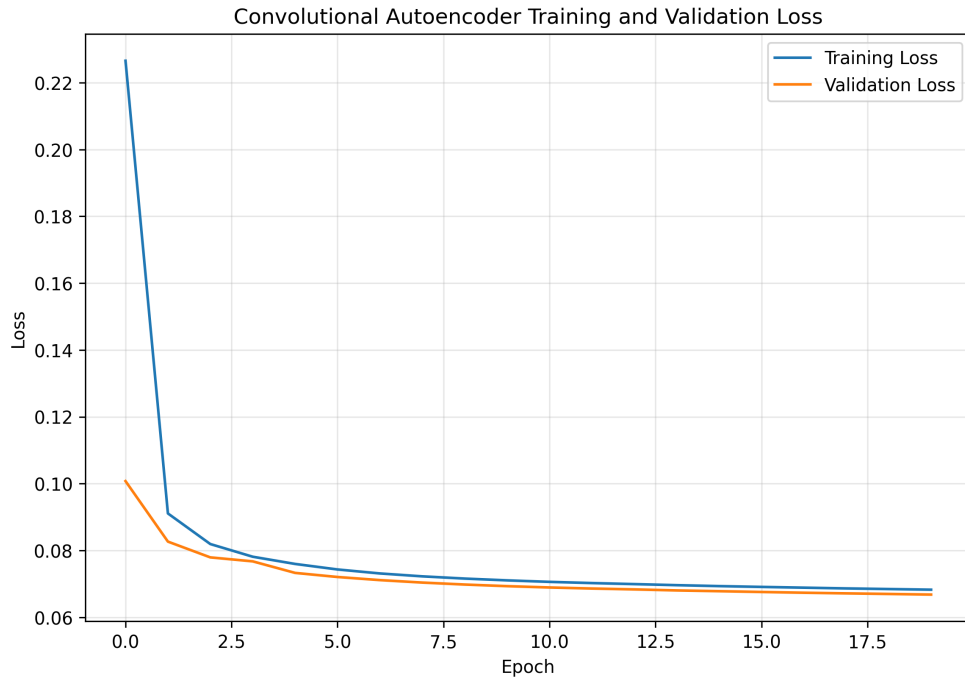


Figure 4: Training and validation reconstruction loss for the convolutional autoencoder.

The final CAE training loss was 0.068556 and validation loss was 0.067027 after 20 epochs. The recorded training time was 661.35 seconds.

7.2 FC-AE versus CAE Reconstruction

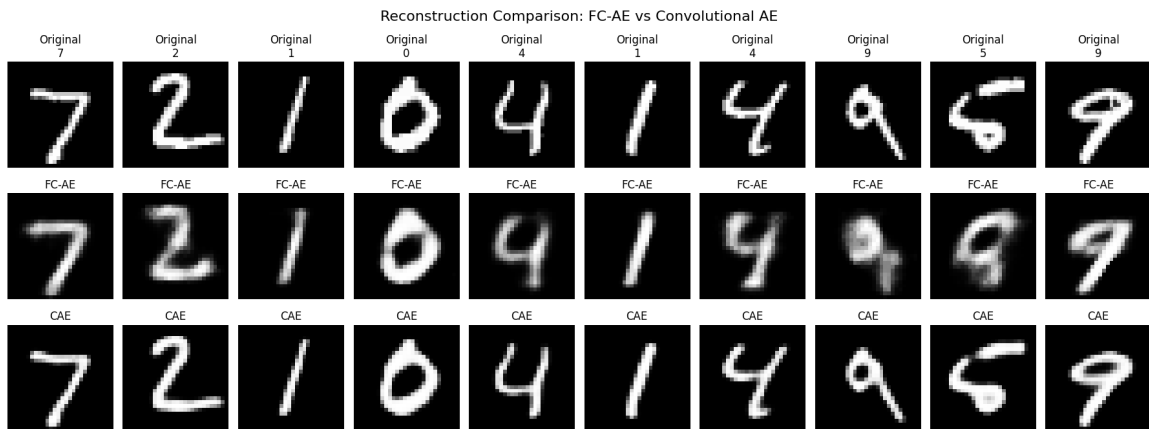


Figure 5: Original image, fully connected reconstruction and convolutional reconstruction.

Inference: The convolutional autoencoder preserves spatial locality through shared convolutional filters. In the executed results, this corresponded to substantially lower reconstruction error and higher SSIM than the fully connected model.

The measured CAE performance was:

Test MSE	0.002537
Test MAE	0.014808
Mean SSIM	0.974641
Parameters	74,497

8. Denoising Autoencoder

A denoising autoencoder receives a corrupted image but uses the corresponding clean image as the reconstruction target:

$$\tilde{x} \rightarrow \text{Encoder} \rightarrow z \rightarrow \text{Decoder} \rightarrow \hat{x}.$$

8.1 Gaussian Noise

Gaussian corruption was generated using

$$\tilde{x} = x + n, \quad n \sim \mathcal{N}(0, \sigma^2),$$

with $\sigma \in \{0.1, 0.2, 0.3\}$ and clipping to $[0, 1]$.

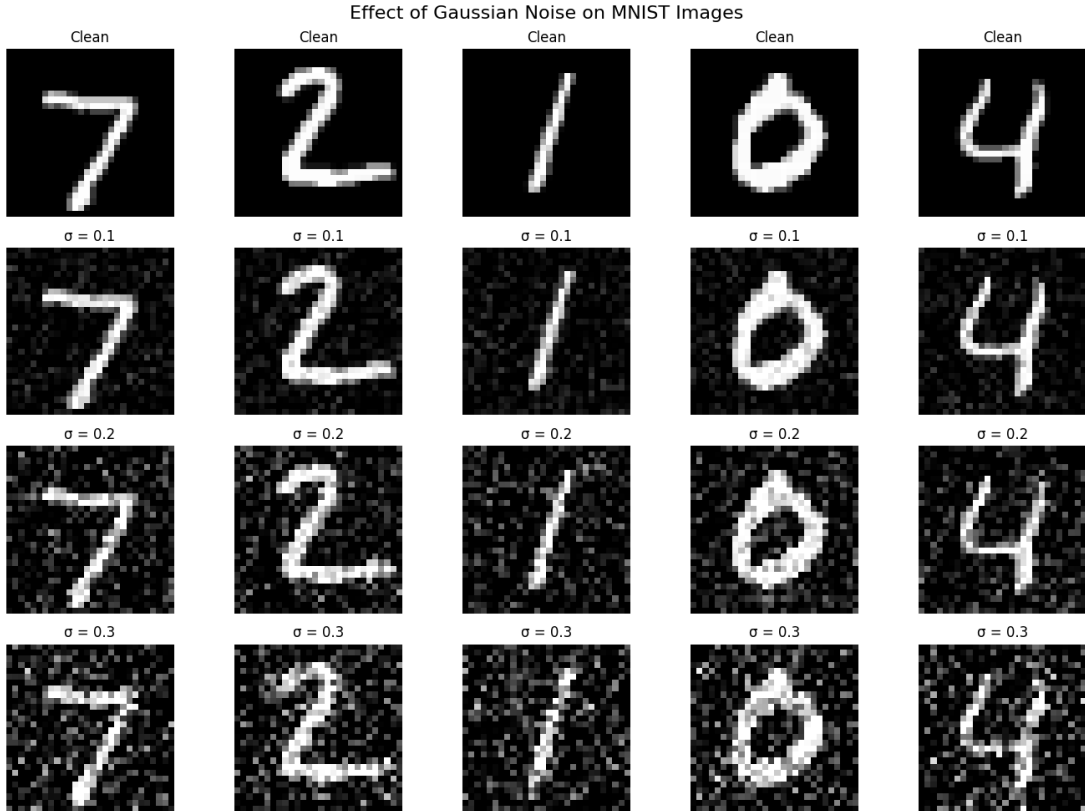


Figure 6: Effect of increasing Gaussian noise on MNIST images.

The denoising CAE was trained using $\sigma = 0.2$ for 20 epochs. Its final training loss was 0.072421 and final validation loss was 0.071459. The recorded training time was 713.39 seconds.

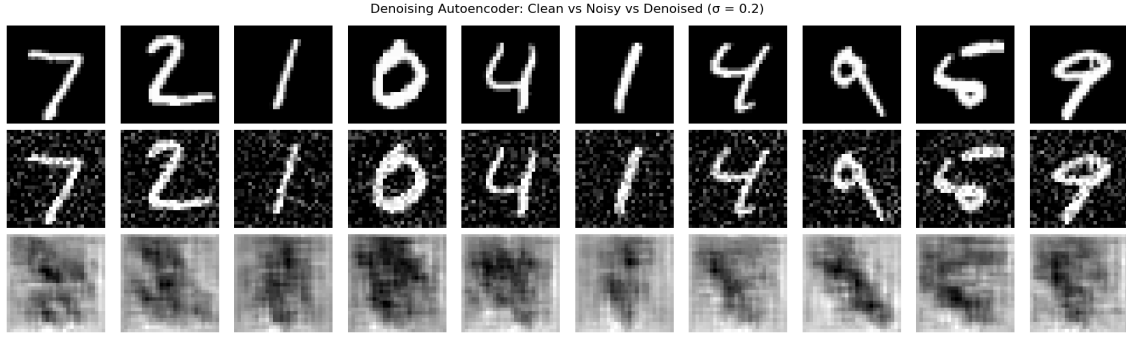


Figure 7: Clean, noisy and denoised MNIST images for Gaussian noise.

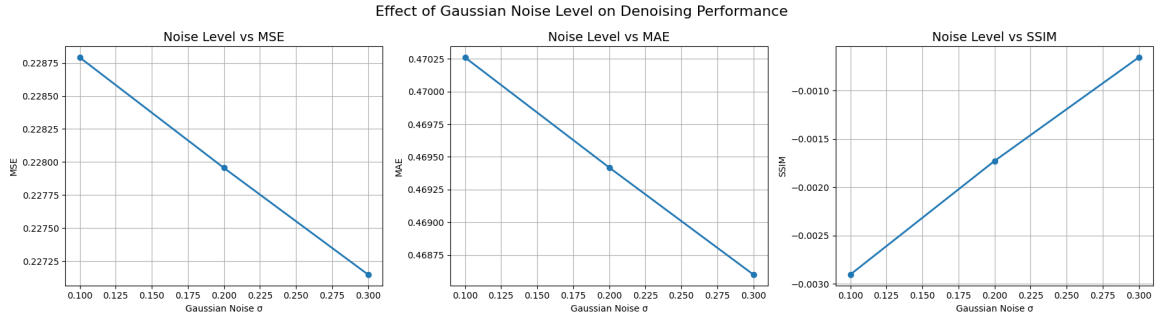


Figure 8: Reconstruction metrics measured at Gaussian noise levels $\sigma = 0.1, 0.2, 0.3$.

Inference: Increasing corruption changes the visual quality of the input images and affects the reconstruction metrics. The denoising model is nevertheless able to recover the major digit structure from corrupted observations. The plotted metric values are those produced by the executed notebook.

8.2 Salt-and-Pepper Noise

A second corruption type was implemented by randomly replacing a proportion of pixels with values near zero or one. The tested probabilities were

$$p \in \{0.05, 0.10, 0.20\}.$$

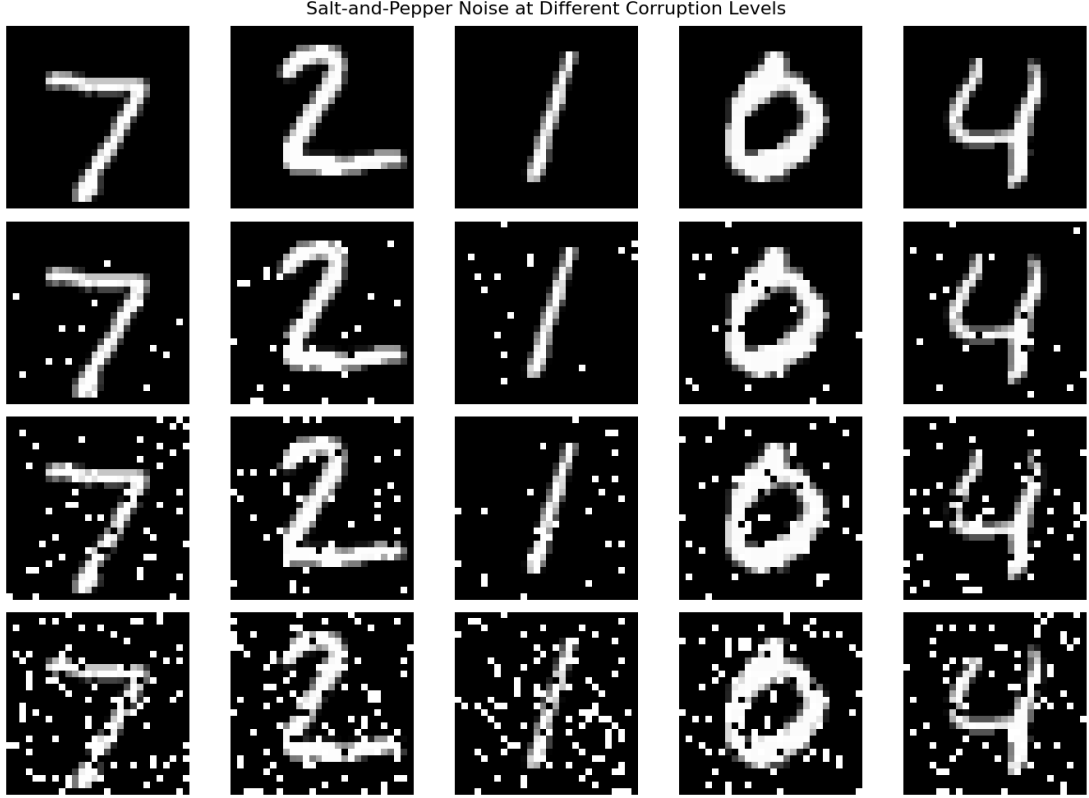


Figure 9: Salt-and-pepper corruption at different probabilities.

For $p = 0.10$, the executed denoising model produced:

$$\text{MSE} = 0.004813, \quad \text{MAE} = 0.020720, \quad \text{SSIM} = 0.946327.$$

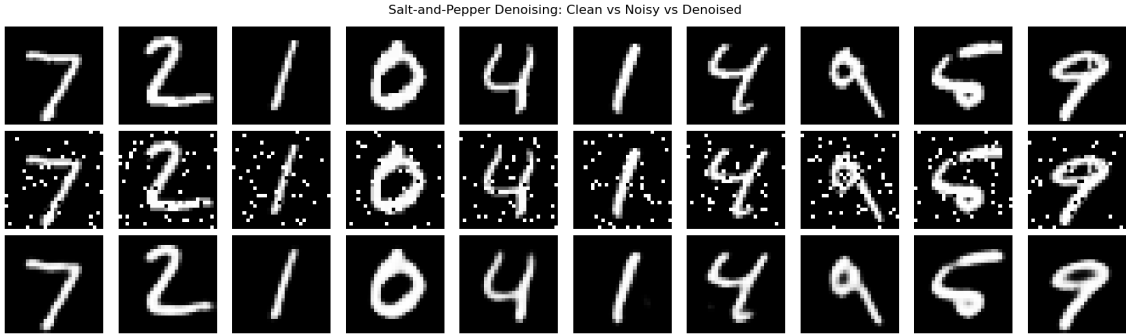


Figure 10: Salt-and-pepper denoising result for $p = 0.10$.

Inference: The denoising network learns a mapping from corrupted images to clean targets, allowing it to recover the dominant digit strokes even when individual pixels are corrupted.

9. Variational Autoencoder

A conventional autoencoder maps an input to a deterministic latent vector. A VAE instead learns a probability distribution:

$$q_{\phi}(z|x) = \mathcal{N}\left(\mu(x), \text{diag}(\sigma^2(x))\right).$$

The encoder produces μ and $\log \sigma^2$. The reparameterization trick was implemented as

$$z = \mu + \sigma \odot \epsilon, \quad \epsilon \sim \mathcal{N}(0, I).$$

The VAE used a two-dimensional latent space:

$$z \in \mathbb{R}^2.$$

The total objective was

$$L_{\text{VAE}} = L_{\text{rec}} + L_{\text{KL}},$$

where

$$D_{\text{KL}}(q_\phi(z|x) \| p(z)) = -\frac{1}{2} \sum_j \left(1 + \log \sigma_j^2 - \mu_j^2 - \sigma_j^2 \right)$$

and the prior was

$$p(z) = \mathcal{N}(0, I).$$

9.1 VAE Training

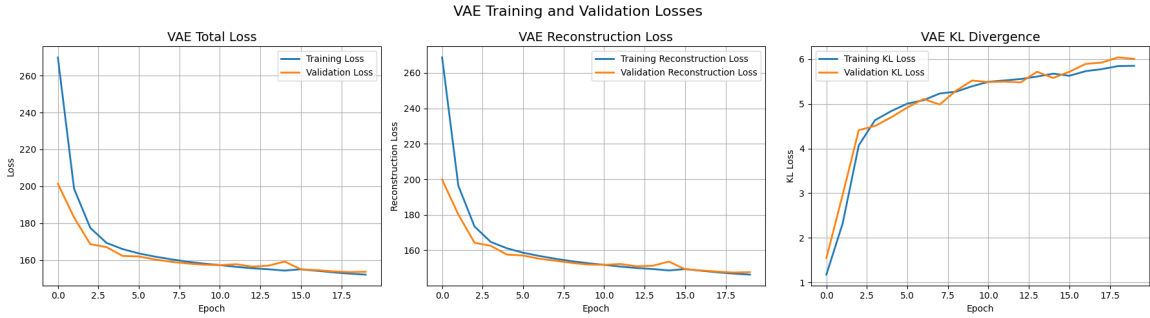


Figure 11: VAE total, reconstruction and KL loss curves during training.

The VAE was trained for 20 epochs with batch size 128, latent dimension 2 and Adam learning rate 10^{-3} . The recorded training time was 638.37 seconds. At the end of training, the reported validation total loss was approximately 153.6936, validation reconstruction loss approximately 147.6846 and validation KL loss approximately 6.0090.

9.2 VAE Reconstruction

The executed test performance was:

Test MSE	0.043134
Test MAE	0.100257
Mean SSIM	0.494859
Mean KL divergence	6.013857
Parameters	485,957

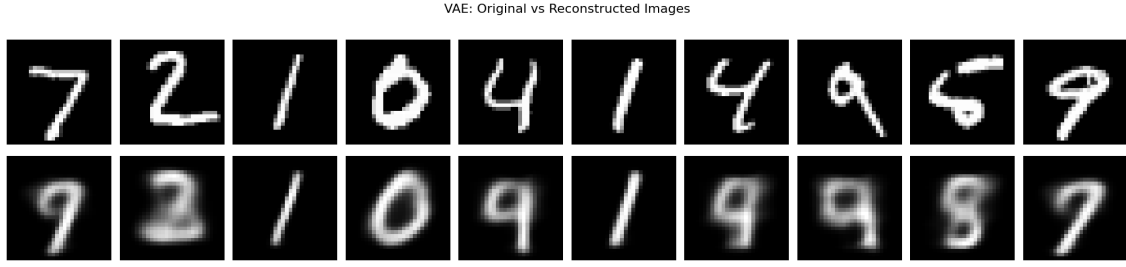


Figure 12: Original images and VAE reconstructions.

Inference: The VAE reconstruction is more probabilistic than the deterministic autoencoder reconstruction. The latent distribution is regularized toward the standard normal prior, and the resulting reconstructions can therefore trade some pixel-level fidelity for a structured latent space suitable for generation.

9.3 Two-Dimensional Latent Space

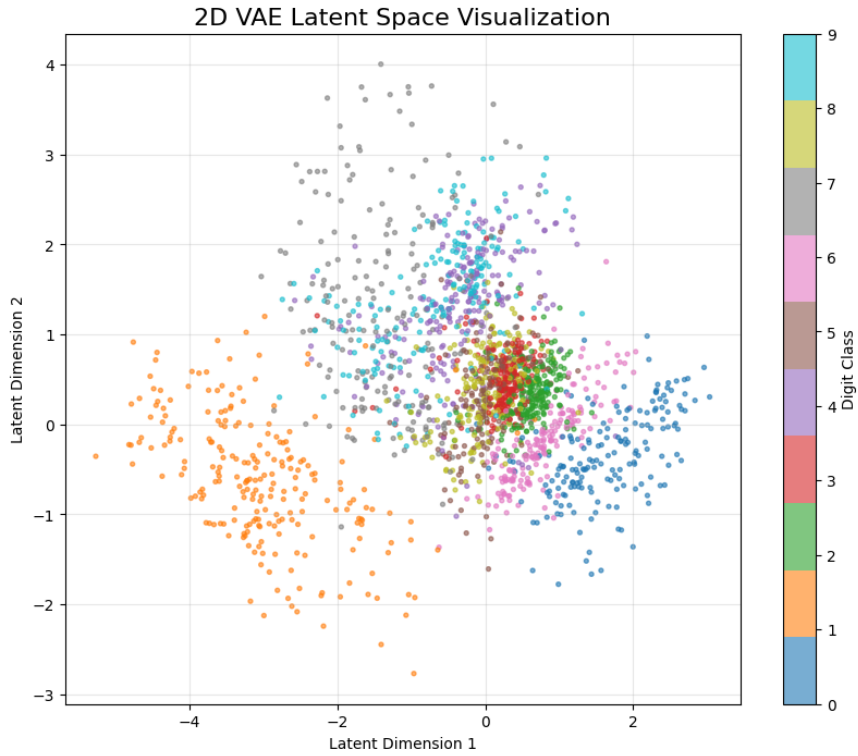


Figure 13: Two-dimensional VAE latent-space visualization, colored by digit label.

The plot represents 2,000 test images using two latent coordinates. Samples of the same digit tend to occupy related regions, although overlap can occur between visually similar digit classes. The labels were used only for visualization and were not provided to the VAE during training.

9.4 Random Image Generation

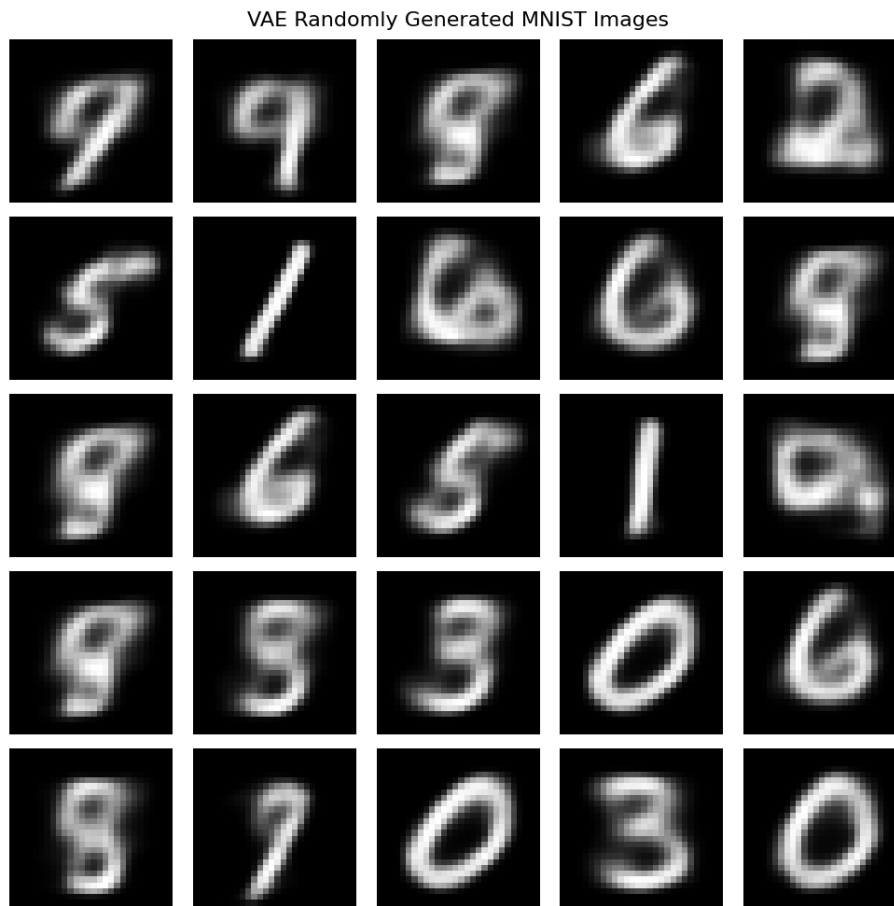


Figure 14: Twenty-five images generated by sampling $z \sim \mathcal{N}(0, I)$.

Inference: Sampling from the learned latent prior allows the VAE to generate new images rather than only reconstructing test images. The generated samples show recognizable digit-like structures with visible variation; some samples are less clear because the sampled latent point may decode to an ambiguous region.

9.5 Latent-Space Interpolation

For two latent vectors z_A and z_B , interpolation was performed using

$$z(\alpha) = (1 - \alpha)z_A + \alpha z_B, \quad \alpha \in [0, 1],$$

with $\alpha = 0, 0.1, \dots, 1.0$.

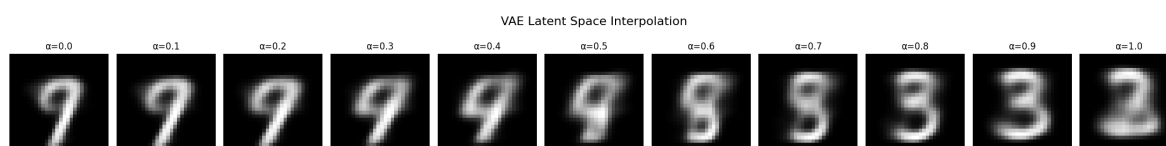


Figure 15: Decoded images along a linear path between two latent points.

Inference: The sequence illustrates how the decoded image changes as the latent point

moves continuously between two encoded samples. Smooth changes indicate that nearby latent representations correspond to related image structures.

10. Reconstruction Error Distribution and High-Error Samples

The per-image reconstruction error was computed as

$$e_i = \frac{1}{D} \sum_{j=1}^D (x_{ij} - \hat{x}_{ij})^2.$$

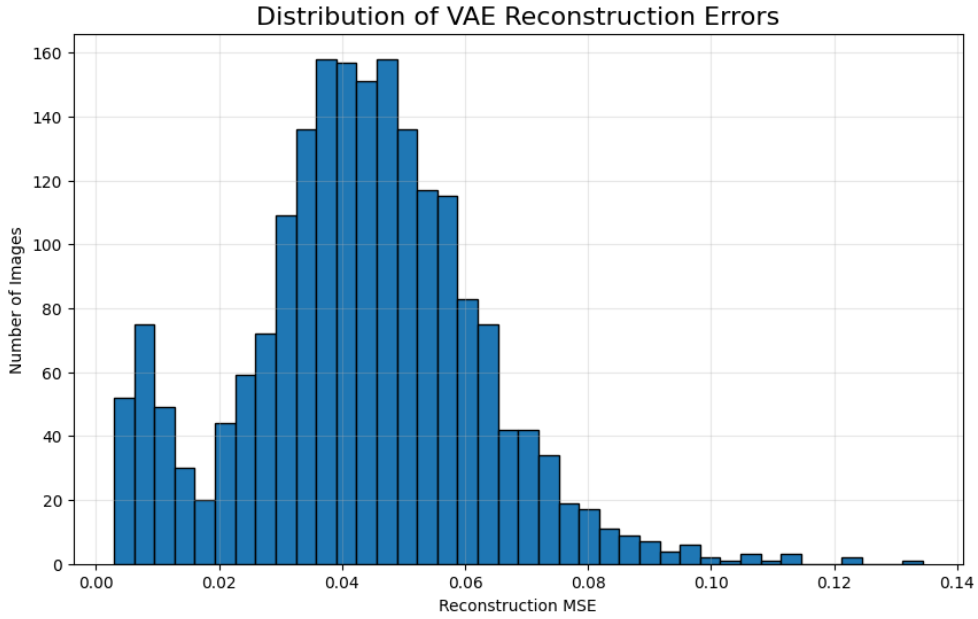


Figure 16: Distribution of VAE reconstruction errors on the test set.

The error distribution shows the range of reconstruction errors across the 2,000 test images. Most images have lower errors, while a smaller group forms the high-error tail.

The five highest-error samples from the executed experiment were:

Rank	Index	Digit	MSE
1	1352	2	0.134376
2	998	8	0.124196
3	1847	5	0.121735
4	1325	8	0.113770
5	352	5	0.113375

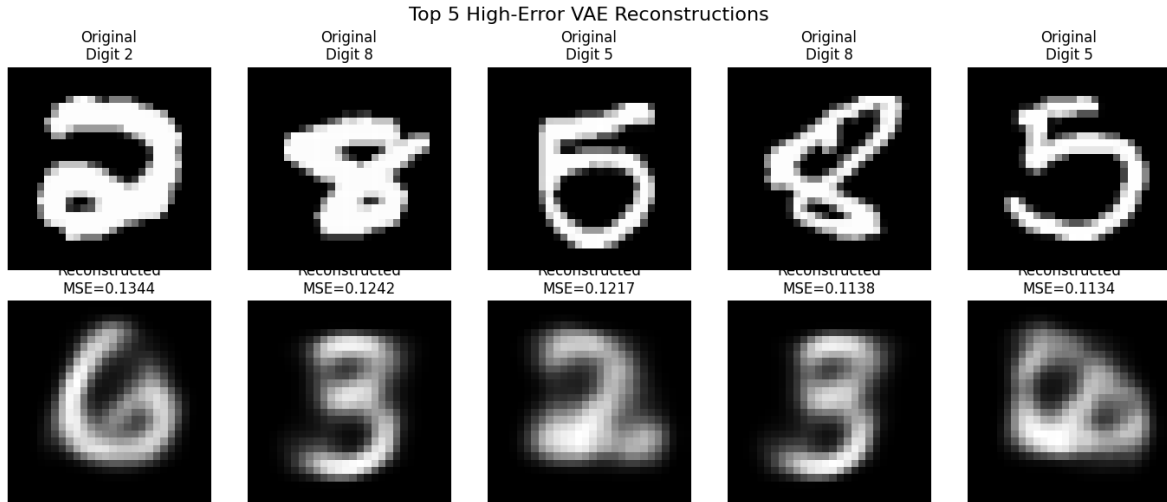


Figure 17: Five test images with the largest VAE reconstruction errors.

The high-error examples contain stroke arrangements that are less well represented by the learned two-dimensional latent space. Their reconstructions are smoother and may alter important local stroke details.

11. Consolidated Model Comparison

The following values were obtained from the executed experiment:

Model	MSE	MAE	SSIM	Parameters	Time (s)
FC Autoencoder	0.022515	0.059259	0.734634	211,040	23.57
Convolutional AE	0.002537	0.014808	0.974641	74,497	661.35
Denoising CAE	0.016222	0.059488	0.642149	74,497	713.39
Variational AE	0.043134	0.100257	0.494859	485,957	638.37

For the VAE, the mean KL divergence was 6.013857. The VAE total objective includes the KL term, so its reconstruction metrics are reported separately from the total training objective.

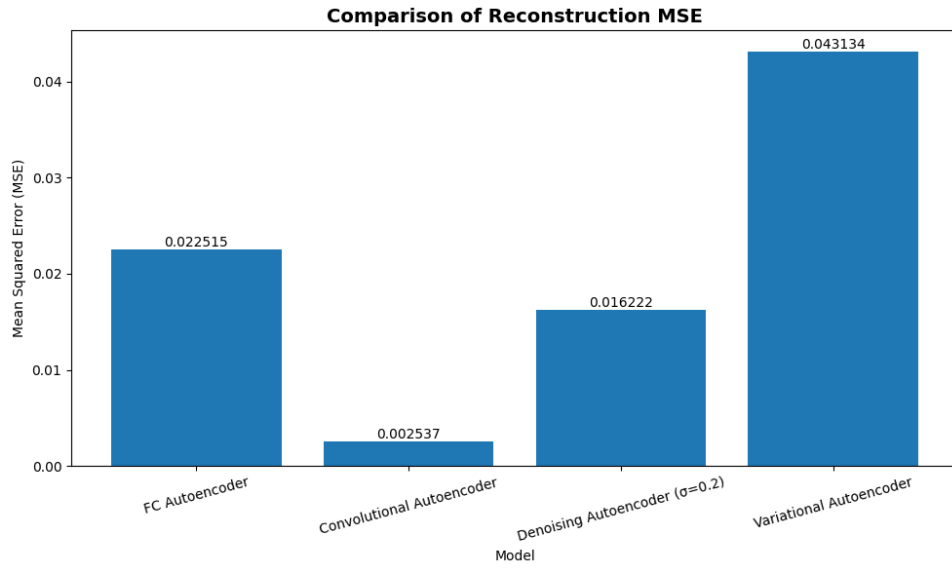


Figure 18: MSE comparison across the four principal models.

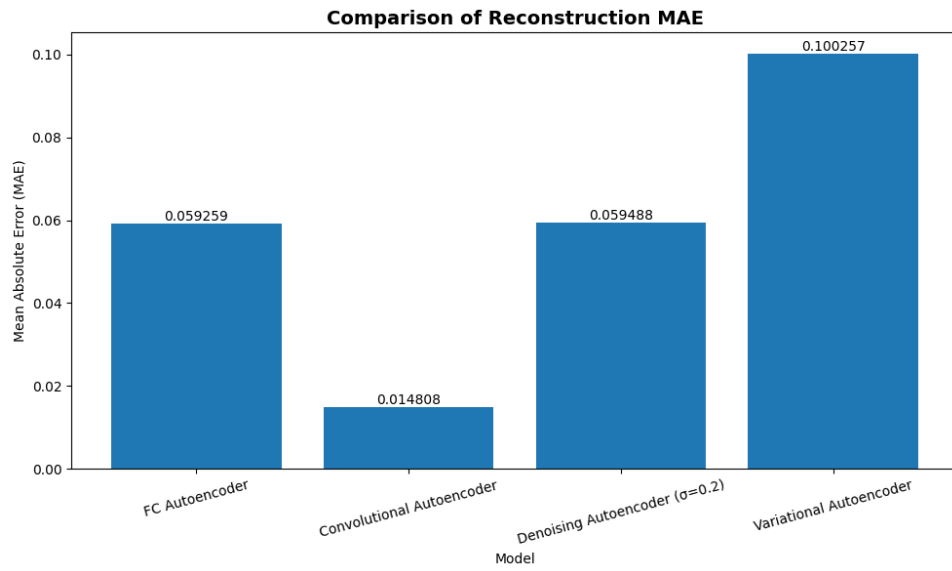


Figure 19: MAE comparison across the four principal models.



Figure 20: SSIM comparison across the four principal models.

Inference: The executed CAE produced the lowest reconstruction MSE and MAE and the highest SSIM among the four reported models. The FC-AE used substantially more parameters than the CAE despite its simpler fully connected structure. The denoising CAE retained the convolutional architecture but its task was more difficult because its input was deliberately corrupted. The VAE had the largest parameter count in this experiment and optimized a probabilistic objective containing both reconstruction and KL terms.

12. Additional Latent-Dimension Study

The additional study evaluated latent dimensions

$$d_z \in \{2, 8, 16, 32\}.$$

The executed values were:

Latent Dimension	MSE	SSIM	Parameters
2	0.048060	0.416966	210,196
8	0.031502	0.624622	210,784
16	0.027183	0.668263	211,568
32	0.028791	0.652213	213,136

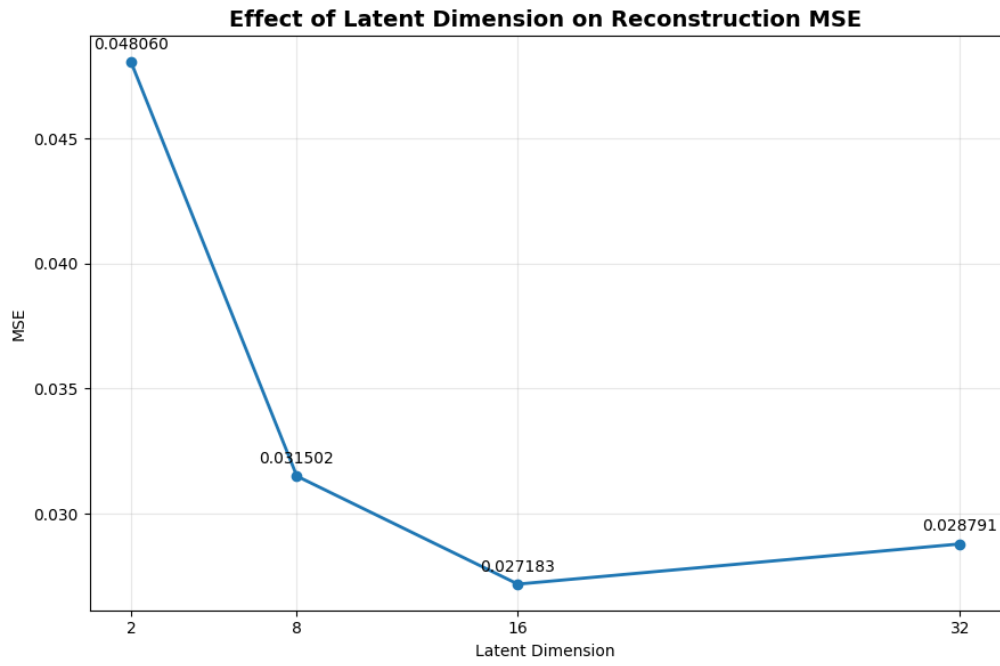


Figure 21: Effect of latent dimension on reconstruction MSE.

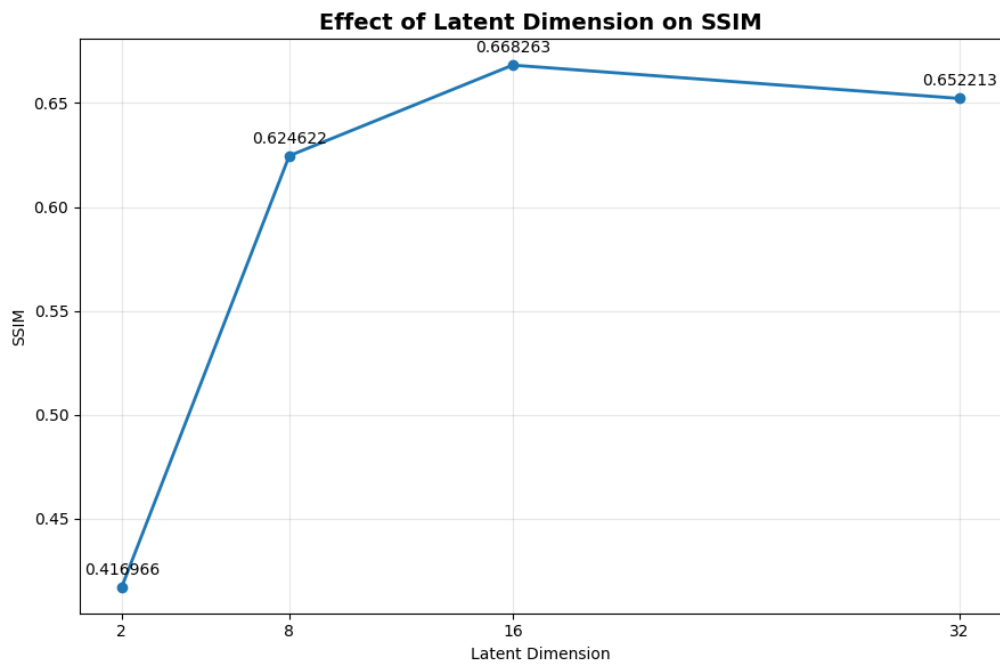


Figure 22: Effect of latent dimension on SSIM.

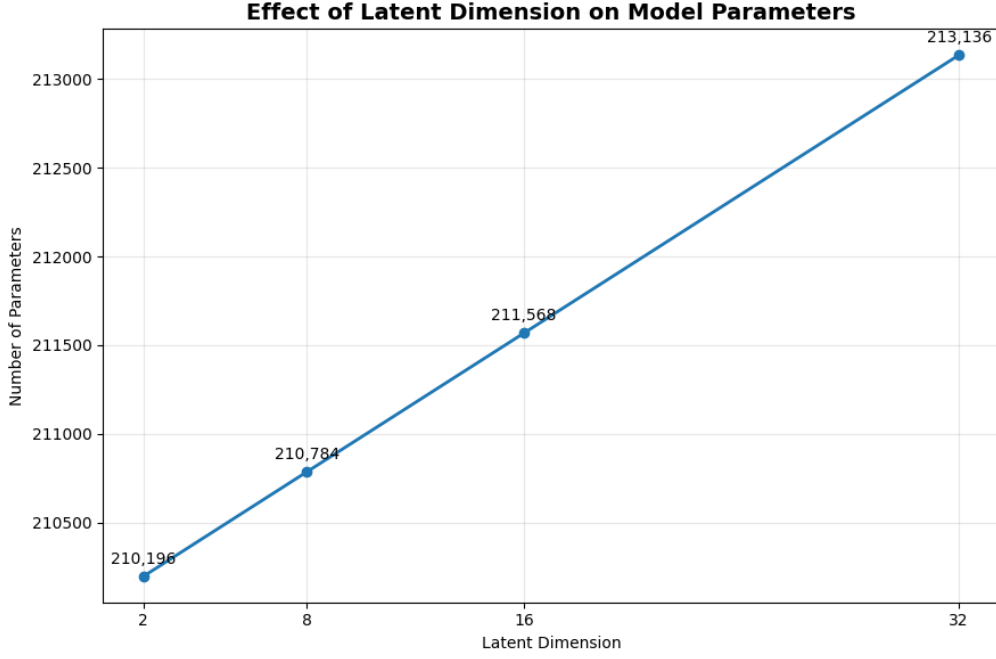


Figure 23: Effect of latent dimension on model parameters.

Inference: Increasing the latent dimension from 2 to 16 substantially reduced MSE and increased SSIM in the executed study. Increasing the dimension further to 32 did not continue the same improvement: MSE increased slightly and SSIM decreased slightly. Thus, the experiment demonstrates a compression–fidelity trade-off rather than an assumption that a larger latent dimension always produces better reconstruction.

13. Required Inferences

1. **Effect of latent dimension:** A very small latent space forces stronger compression and can lose image information. In the executed study, increasing the latent dimension from 2 to 16 improved reconstruction metrics, while the change from 16 to 32 did not provide further improvement.
2. **Fully connected versus convolutional reconstruction:** The CAE preserves spatial locality through convolution and therefore reconstructed the MNIST images with substantially lower error than the FC-AE in this run.
3. **Effect of Gaussian noise:** Larger Gaussian noise levels visibly corrupt the digit structure and change the measured reconstruction quality.
4. **Denoising ability:** Training with noisy input and clean targets allows the denoising network to learn recovery of the dominant digit structure.
5. **VAE latent distribution:** The VAE learns a probabilistic representation regularized toward a standard normal prior, enabling sampling from the latent space.
6. **Two-dimensional latent structure:** The two latent coordinates form a visual representation in which related digit samples tend to occupy related regions, with overlap between some classes.

7. **Generation quality and diversity:** Random latent samples produce varied digit-like images, although some generated images can be ambiguous.
8. **Interpolation smoothness:** Linear interpolation in latent space produces a sequence of decoded images that changes progressively between the selected endpoints.
9. **Reconstruction error versus visual quality:** High numerical reconstruction error is associated with visually difficult samples whose stroke patterns are less faithfully reproduced.
10. **Deterministic versus probabilistic latent representations:** The conventional autoencoder maps an image to a single deterministic vector, whereas the VAE represents a distribution through μ and $\log \sigma^2$ and samples from it.

14. Discussion Questions

1. **What is an autoencoder?** An autoencoder is a neural network trained to reconstruct its input through an internal latent representation.
2. **What is the purpose of the encoder?** The encoder transforms the input image into a compact latent representation.
3. **What is the purpose of the decoder?** The decoder maps the latent representation back to the image space.
4. **What is a latent representation?** It is the internal compressed representation learned between the encoder and decoder.
5. **Why is it usually smaller than the input?** A smaller latent space forces the network to retain the most useful information needed for reconstruction.
6. **What happens when the latent dimension is too small?** Excessive compression causes information loss and poorer reconstruction.
7. **What happens when the latent dimension is increased?** More information can be retained, although the additional capacity does not necessarily improve reconstruction indefinitely.
8. **Why can a convolutional autoencoder preserve spatial information more effectively?** Convolutional filters operate on local neighborhoods and preserve spatial relationships between pixels.
9. **How does a denoising autoencoder differ from an autoencoder?** A denoising autoencoder receives a corrupted input but is trained against the original clean image.
10. **Why is the clean image the target in denoising?** The clean target provides the desired signal that the network must recover from the corrupted observation.
11. **What happens when noise is increased?** The input becomes more corrupted and reconstruction becomes more difficult.
12. **Why are MSE, MAE and SSIM useful?** MSE and MAE quantify pixel-level errors, while SSIM evaluates structural similarity.
13. **What is a VAE?** A VAE is an autoencoder that learns a probability distribution over latent variables.

14. **How does a VAE differ from a conventional autoencoder?** A conventional autoencoder uses a deterministic latent vector, whereas a VAE learns distribution parameters and samples latent vectors.
15. **Why does a VAE learn a distribution?** A structured continuous distribution makes latent sampling and generation possible.
16. **What is the reparameterization trick?** It expresses a random sample as $z = \mu + \sigma\epsilon$ so that gradients can be propagated through the sampling operation.
17. **Why is ϵ sampled from a standard normal distribution?** It separates the randomness from the learned parameters μ and σ .
18. **What is the role of KL divergence?** It regularizes the learned latent distribution toward the chosen prior.
19. **What prior was used?** The standard normal distribution $\mathcal{N}(0, I)$.
20. **Why is a two-dimensional latent space useful?** It can be directly visualized as a two-dimensional plane.
21. **What does latent-space interpolation demonstrate?** It demonstrates how decoded outputs change along a continuous path between two latent representations.
22. **Why can a VAE generate new images?** New latent vectors can be sampled from the learned prior and passed through the decoder.
23. **Why might generated images be unclear?** A sampled point may decode to an ambiguous region of the learned latent distribution.
24. **What is the difference between reconstruction and generation?** Reconstruction decodes a latent representation obtained from an input image; generation starts from a newly sampled latent vector.
25. **Why should test images remain unseen during training?** Unseen test images provide a measure of how the learned model generalizes beyond its training examples.

15. Conclusion

The experiment demonstrated a complete image representation pipeline using four autoencoder variants. The fully connected autoencoder established deterministic encoding and reconstruction, while the convolutional autoencoder preserved spatial locality and achieved substantially improved reconstruction metrics in the executed MNIST experiment. The denoising autoencoder showed that a model can learn to recover clean digit structure from deliberately corrupted inputs. The VAE introduced probabilistic latent representations, KL regularization, latent-space visualization, random generation and interpolation.

The additional latent-dimension study demonstrated the trade-off between compression and reconstruction fidelity. Overall, the experiment covered deterministic reconstruction, convolutional image representation, denoising and probabilistic generation within a common MNIST-based framework.

16. References

1. Ian Goodfellow, Yoshua Bengio and Aaron Courville, *Deep Learning*, MIT Press, 2016.
2. Diederik P. Kingma and Max Welling, “Auto-Encoding Variational Bayes,” ICLR, 2014.
3. Pascal Vincent et al., “Stacked Denoising Autoencoders: Learning Useful Representations in a Deep Network with a Local Denoising Criterion,” *Journal of Machine Learning Research*, 2010.
4. Yann LeCun, Corinna Cortes and Christopher J. C. Burges, “MNIST Handwritten Digit Database.”
5. TensorFlow Documentation: <https://www.tensorflow.org>
6. Keras Documentation: <https://keras.io>
7. scikit-image Documentation: <https://scikit-image.org>